

NOUHAYLA DARBALI

Casablanca, Morocco — +212 679105444 — darbalinohayla@gmail.com

 nouhayladarbali.vercel.app —  linkedin.com/in/nouhayladarbali —  github.com/DarbaliNouhayla

 huggingface.co/darbalinohayla —  kaggle.com/shiranaihito

AI Engineering student (3rd year, EHTP) building end-to-end ML systems across Computer Vision, LLMs, and Multi-Agent architectures. Hands-on experience fine-tuning models (LoRA, SegFormer, YOLO), deploying async FastAPI services and models on Hugging Face, and shipping AI products during internships. Seeking a PFE internship in AI/ML Engineering starting early 2027; open to remote, hybrid, or on-site.

Education

Ecole Hassania des Travaux Publics (EHTP)

Master of Engineering in Computer Science

Sep. 2024 – May 2027

Casablanca, Morocco

CPGE Ibn Taher (MP)

Preparatory Classes: Applied Mathematics, Physics, and Computer Science

Sep. 2022 – July 2024

Errachidia, Morocco

High School Diploma (Baccalaureate)

Mathematical Sciences – Mention Très Bien (Honors)

2022

Zagora, Morocco

Professional Experience

Computer Vision & AI Engineering Intern

Jul. 2026 – Sep. 2026

Civil Infrastructure AI Lab / CID

Casablanca, Morocco

- Engineered semantic segmentation pipelines for structural defect detection, benchmarking CNNs (U-Net, DeepLabV3+) against transformer-based models (SegFormer, SAM3) and YOLOv11; SegFormer-B2 achieved the best results.
- SegFormer-B2 reached **0.73 IoU**, **0.79 Dice**, **0.83 precision** and **0.86 recall** on the held-out test set.
- Annotated defect data in Roboflow and curated it in FiftyOne, adding 10% synthetic negatives and Albumentations augmentation to target false positives; trained with custom PyTorch Focal + Dice loss.

Software & AI Engineering Intern

Jul. 2026 – Sep. 2026

Financial Intelligence & TPRM Platform / Kainos Technologies

Remote

- Built an extensible async OSINT ingestion engine scraping **25+ financial sources** (Playwright, httpx), using **Adapter, Strategy, Factory** patterns so new sources plug in with minimal code, with **PostgreSQL 18** (asyncpg) and TTL deduplication.
- Architected an orchestrated **Multi-Agent System (MAS)** with **Groq API (Llama 3.1 8B)** and **FastAPI** for automated supplier risk profiling, market trend analysis, and CVE vulnerability enrichment.

Full-Stack Software Development Intern

Jul. 2025 – Aug. 2025

Provincial Administration

Zagora, Morocco

- Developed an HR Management System using **Django** and **MySQL**, featuring role-based security (RBAC) and automated PDF balance generation (**ReportLab**) for 100+ personnel.

Technical Projects

ResearchPal AI: Agentic RAG Ecosystem | Python, LangChain, Unsloth, HuggingFace, ChromaDB

2025

- Built an agentic RAG research assistant where the LLM autonomously calls **MCP** tools (paper search, paper details) to fetch live papers from the **ArXiv API** and answer questions from them.
- Fine-tuned **Llama 3.2-3B** with **LoRA** (0.42% of parameters) on SciQ (13.7K Q&A) to explain science to students; **Unsloth** 4-bit training cut VRAM by 60% vs. standard LoRA (~15 GB to ~6 GB), fitting a free T4 GPU.

Traffic AI Morocco & ADAS Safety System | Python, YOLOv8, ByteTrack, OpenCV, OSRM

2026

- Designed an urban tracking system integrating **YOLOv8**, **ByteTrack**, custom Arabic **OCR**, and **OSRM** route optimization with driver anomaly detection.

Predictor Engine: Football Match Prediction | Python, Scikit-learn, hmmlearn, Pandas

2024

- Benchmarked **6 algorithms** (SVM, Random Forest, KNN) for football match outcome prediction, with feature engineering and normalization pipelines (Pandas, Scikit-learn) on 10 years of data.
- Modeled dynamic team performance with **Hidden Markov Models (HMM)** to capture latent team states.

SecurAI: Voice Cloning & Deepfake Security (Team Project) | Python, PyTorch, Beatrice v2, FastAPI

2026

- Built the voice cloning pipeline on the **Beatrice v2** engine with targeted convolution modifications, served through a **FastAPI** service.
- Contributed to a team platform covering multi-modal **deepfake detection** (spatial, temporal, acoustic anomalies) and simulated agentic attacks (**AACA**) to test defensive resilience.

Technical Skills & Languages

Programming: Python (Asyncio), C++, SQL, JavaScript, HTML/CSS

AI / Vision / Agents: PyTorch, ONNX, SegFormer, SAM3, YOLOv8/v11, Multi-Agent Systems (MAS), Groq API (Llama 3.1), LangChain, ChromaDB, OpenCV, FiftyOne, Roboflow, Scikit-learn, Pandas, HMM (hmmlearn)

Backend & Frameworks: FastAPI, Django, PostgreSQL (asyncpg), Playwright, httpx, React, Three.js

DevOps, IoT & Architecture: Docker, Git, Linux, GCP, Hugging Face Spaces, Vercel, MySQL, IoT, Design Patterns

Spoken Languages: Arabic (Native), French (Professional), English (Fluent), Italian (Intermediate)